

Justin Jasper

(770) 480-8093 | jtjasper@stanford.edu | <https://www.linkedin.com/in/justin-jasper>

EDUCATION

Stanford University

Master of Science in Computer Science (Concentration: Artificial Intelligence)

Stanford, CA

June 2026

Stanford University

Bachelor of Science in Bioengineering

Stanford, CA

June 2025

EXPERIENCE

Centre for Artificial Intelligence and Robotics

Biomedical AI Intern

Hong Kong, China

June 2025 - September 2025

- Developed and deployed a novel full-stack landmark detection model for cardiac ultrasound images to assist non-specialist physicians in accurate diagnosis (Pytorch, Python, React, PostgreSQL).
- Collaborated with the Emergency Department at Prince of Wales Hospital to test the cardiac ultrasound model and integrate it into clinical workflows for point-of-care ultrasound
- Designed a hallucination-resistant LLM architecture leveraging biomedical knowledge graphs to support clinical decision-making and treatment plan formulation based on patient medical history and clinical data (Python, Neo4j)

MITRE Corporation

Computational Biology Intern

Mclean, VA

June 2024 - December 2024

- Developed an LLM-driven retrieval augmented generation system to extract and summarize key insights from complex scientific literature, streamlining knowledge access for synthetic biology workflows (Used scientific python, LangChain, LlamaIndex, Kubernetes).
- Constructed the BioNet Domain-Specific Language, an open-source programming language designed to enable extensive scaling of synthetic biology experiments and industrial workflows.

Omniwear AI

CoFounder, CEO

Palo Alto, CA

December 2022 - December 2023

- Led the development of a virtual try-on platform for e-commerce powered by computer vision and generative AI.
- Participated in Launchpad (Stanford's premier graduate accelerator) Spring '23 and Entrepreneur First's NYC unicorn training program.
- Piloted the tech with Abercrombie & Fitch, Revolve, and Everlane, where we showcased an average return rate reduction of over 50%.

PROJECTS

CV Image Processing for CT and Ultrasound-based Tumor Detection

- Built a Python/OpenCV pipeline to segment and reconstruct 3D lung CT scans for cancer diagnosis, training an Attention U-Net (TensorFlow) for malignant/benign classification and applying K-means clustering to segment pulmonary abnormalities.

Python-Based Probabilistic Modeling of Biological Systems

- Built Python computational models simulating molecular systems — including ion channel activation, protein-ligand binding, and polymer conformations — using Boltzmann statistical mechanics to predict multi-state behaviors and system energetics.

Computational Analysis of NMDA Receptor Antagonist Binding for Chronic Pain Therapeutics

- Used SwissDock and AutoDock Vina to model NMDA receptor antagonist binding interactions, optimizing docking workflows and quantifying binding affinities to evaluate drug repurposing candidates for chronic pain.

Synthetic Biosensor Development for Pathogen Detection

- Engineered a novel Salmonella biosensor using a SynNotch receptor system with an anti-Salmonella scFv transfected into HEK293 cells, validated via flow cytometry, Western blotting, PCR, and gel electrophoresis.

Computer Vision for Meal Nutritional Analysis

- Built a full-stack web app (Flask + HTML) for real-time food nutritional analysis from user-uploaded images, powered by a ResNet18 CNN with an object detection pipeline for food localization and classification.
- <https://github.com/justinjasper/cv-nutritional-analysis>

SKILLS

- Proficient in Python (NumPy, Pandas, SciKit-learn, TensorFlow), C++, C, MATLAB, git, UNIX, LLM training & deployment (LangChain, LlamaIndex, Kubernetes), retrieval augmented generation (RAG), neural networks
- Proficient in Spanish, with strong cross-cultural communication skills.